MOSA: Revision Summary

Cai, Apolinário et al., *Synthetic augmentation of cancer cell line multi-omic datasets using unsupervised deep learning*. Nature Communications 15:10390 (2024)

1523	25.8%	+32.7%	+34.9%	-39.2%
cancer cell lines	had all 7 omics	multi-omic profiles	CRISPR screens	parameters (asym. VAE)

1. Problem and Idea (সমস্যা ও মূল ধারণা)

Problem: Cancer cell lines (ক্যান্সার কোষের ল্যাব-সংস্করণ) have 7 kinds of **omics** data, but most cell lines miss several of them. Experiments are costly, so only **25.8%** have all seven. (DepMap data: *heterogeneity* বৈচিত্র্য, *sparsity* বিরলতা, missing values.)

Solution: MOSA = **M**ulti-**O**mic **S**ynthetic **A**ugmentation, an **unsupervised** (label ছাড়াই শেখে) **conditional multi-view VAE** (PyTorch) that learns one shared summary of a cell line from all its available omics and then **generates the missing omics**.

> **BANGLA INTUITION**
> ১০০ জন ছাত্রের কারো গণিত নেই, কারো পদার্থ নেই। বাকি বিষয়ের নম্বর দেখে হারানো নম্বর অনুমান করা যায়। এখানে ছাত্র = ক্যান্সার কোষ, বিষয় = omics।

The 7 omics (সাতটি স্তর)

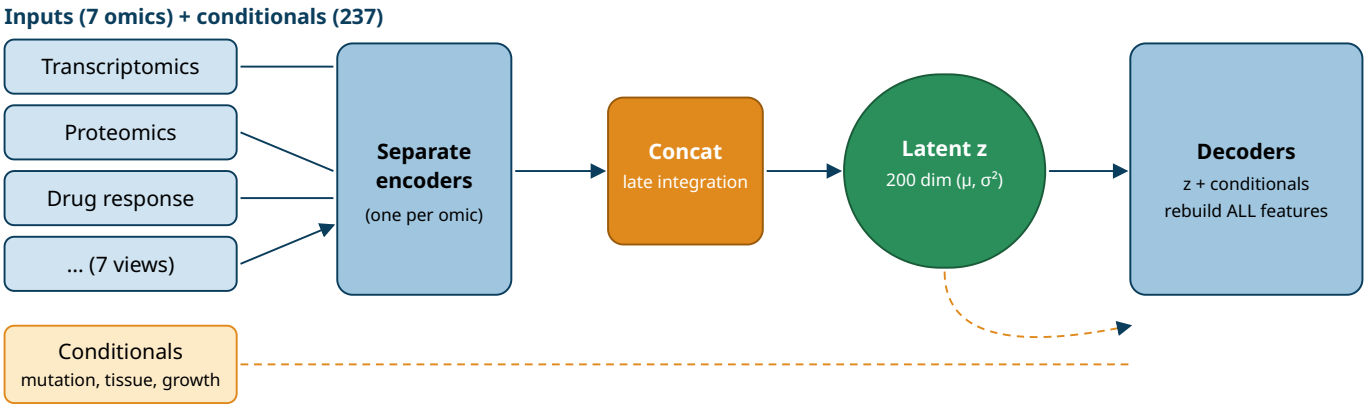
Omics	Measures (কী মাপে)	Features	Cell lines (Fig 1a)
Genomics	Mutations, copy number (জিনের গঠন)	CN: 777	CN: 1,198
Transcriptomics	Gene expression / RNA (কোন জিন কতটা সক্রিয়)	15,278	1,240
Proteomics	Protein abundance (প্রোটিনের পরিমাণ)	4,922	949
Methylomics	DNA methylation (জিনের on/off ট্যাগ)	14,608	906
Metabolomics	Metabolites (কোষের রাসায়নিক পদার্থ)	225	900
Drug response	IC50 sensitivity (ওষুধে সাড়া)	810	993
CRISPR-Cas9	Gene essentiality (জিন বন্ধ করলে কোষ মরে কি না)	17,931	992

> **EXAM POINT**
> Vertical integration = integrating different data types on the **same samples**. A cell line is included only if it has **at least 2** omics. Existing linear methods (e.g. MOFA) cannot capture non-linear relations; earlier VAEs were built for single-cell count data (scRNA-seq, scATAC-seq) so they do not fit bulk multi-omics.

2. Architecture (গঠন)

VAE in one line (এক লাইনে VAE)

Encoder (সংকোচক) compresses data into a **latent space** (লুকানো সংক্ষিপ্ত উপস্থাপনা, "কোষের পরিচয়পত্র"); **Decoder** (পুনর্গঠক) rebuilds the data. A good summary means the model understood the data. The encoder outputs a **mean (μ)** and **variance (σ^2)**; z is sampled from this Gaussian.



Conditionals go to encoders (before) AND decoders (after z)

Four key design choices (চারটি বিশেষ কৌশল)

Technique	What it does (English)	বাংলা intuition
Conditionals (n=237)	Binary/continuous context: driver mutations, gene fusions, tissue of origin, MSI-high, growth rate. Concatenated to encoder inputs *and* to z before decoding. Mutations enter as conditionals (not a separate view) because genetic background conditions all other omics.	রোগীর বয়স, পেশা জানলে রিপোর্ট বোঝা সহজ। কোষের "জেনেটিক ব্যাকগ্রাউন্ড" = প্রসঙ্গ।
Asymmetrical VAE	Encoder gets only the most variable features (Gaussian mixture, k=2, std-dev threshold: 1.122 for RNA, 0.064 for methylation; never-essential CRISPR genes dropped). Decoder rebuilds **all** features. Cuts parameters by **39.2%** with low error.	যে জিন সব কোষে একই, তার থেকে নতুন কিছু শেখা যায় না। বাদ দিলে মডেল হালকা হয়।
View dropout	Zeroes an **entire omic layer** at random (not single features). Stops big datasets (e.g. CRISPR 17,931 features) from dominating; forces cross-omic learning; improves reconstruction and tissue clustering.	পরীক্ষায় এক বিষয়ের বই লুকিয়ে দিলে বাকি বিষয় থেকে শিখতে হয়।
Contrastive loss	Pulls same-tissue cell lines together, pushes different tissues apart in latent space.	সমগোত্রীয়রা কাছে, ভিন্নরা দূরে।

> **NOTE**
> Late integration: separate encoders, then concatenation. Product-of-experts was tried but **concatenation gave the lowest reconstruction loss**. Paper inconsistency: text says best view_dropout = 0.5, Table 1 lists 0.3.

3. Loss Function (ক্ষতি ফাংশন)

$$L_{total} = L_{recon} + \lambda \cdot L_{KL} + \alpha \cdot L_{contrastive} \quad (\lambda = 0.0001, \alpha = 0.005)$$

Term	Formula / meaning	সহজ কথায়
L_{recon}	$\Sigma_d l_d$, with l_d = **MSE** per dataset d. Missing values are masked (excluded from the loss).	কপিটা কতটা নির্ভুল?
L_{KL}	**KL divergence** (দুটি বণ্টনের দূরত্ব) between learned N(μ, σ^2) and standard normal N(0,1). Regularises latent space.	latent space যেন মসৃণ ও সুসংগঠিত থাকে।
$L_{contrastive}$	$[m_{pos} - s_p]_+ + [s_n - m_{neg}]_+$, s = cosine similarity; positive pair = same tissue. Margins m_{pos}=0.85, m_{neg}=0.15.	একই tissue হলে similarity ≥ 0.85, ভিন্ন হলে ≤ 0.15 চাই।

MSE and Pearson's r (worked example, ধাপে ধাপে)

$$MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$$

Student	Actual y	Predicted ŷ	Error	Error²
A	80	78	2	4
B	60	65	−5	25
C	90	90	0	0

MSE = (4 + 25 + 0) / 3 = **9.67**; RMSE = √9.67 ≈ **3.11**.

> **WHY SQUARE? (কেন বর্গ?)**
> (1) +5 ও −5 যেন বাতিল না হয়। (2) বড় ভুলকে বেশি শাস্তি: error 10 হলে 100, কিন্তু error 2 হলে 4। (3) smooth, তাই gradient descent সহজ। **MSE যত কম তত ভালো; Pearson's r যত বেশি (max 1) তত ভালো।** Outlier থাকলে MSE অনেক বাড়ে।

4. Baselines (তুলনামূলক পদ্ধতি)

Mean imputation

Fill each missing value with that feature's **mean**. Example: marks 80, 70, 90, 60 give mean 75, so a missing student gets 75. Ignores the sample's own information, so it forms horizontal stripes in scatter plots (Fig 2e). Serves as the **minimum baseline**.

MOFA (Multi-Omics Factor Analysis)

A **linear** factor model, the multi-omics version of PCA: **Data ≈ Z × W** (Z = latent factors, W = loadings/weights). Missing omic is predicted as Z × W_{omic}. Used 97 factors. Limitation: only linear relationships; no mixed Gaussian/Bernoulli view.

Others: MOVE (VAE-based; poor here), **mixOmics/DIABLO** (correlation-based, not generative, clustering only). Of **16** methods tried, only these 3 handle all 7 omics. For a 2-omic benchmark (RNA + drug) they added iClusterPlus, JAMIE, scVAEIT, moCluster.

5. Results (ফলাফল)

Test	Result
10-fold CV, drug response	mean feature Pearson r = **0.65**
10-fold CV, CRISPR-Cas9	mean feature r = **0.35** (beats supervised baseline 0.25)
New GDSC drug dataset (unseen in training; 32,659 IC50s)	r = **0.87**, MSE = 1.36
Transcriptomics for 272 lines with *no* RNA in training	mean r = **0.90**
Proteomics: ~32% of matrix filled	Correlation with CCLE proteomics preserved; PPI recall improved with GLM
CRISPR: genetic association power	More significant hits (Fig 3c); skew correlation r = 0.52 (likely underestimated)

Benchmark on unseen drug data (Fig 2)

Model	Pearson r	Spearman ρ	MSE
MOSA	**0.87**	**0.81**	**1.36**
MOFA	0.86	0.79	1.42
Mean imputation	0.77	0.67	2.18
MOVE	0.33	0.36	15.79

EXAM POINT

Biological sanity checks: **BRAF** dependency recovered in BRAF-mutant lines (mostly V600E); **FLI1** dependency recovered in **FLI1-EWSR1** fusion lines. Discrepancies between MOSA and data exposed likely bad measurements (e.g. trametinib; venetoclax and anti-apoptotic drugs lack good biomarkers). Tissue-of-origin clustering is best using **transcriptomics + drug response** because these are strongly tissue-structured; proteomics and metabolomics are looser. Adding more omics to MOSA improves over other methods.

6. Explainability with SHAP (ব্যাখ্যাযোগ্যতা)

Black box (কালো বাক্স) problem: deep nets do not say why. **SHAP** = SHapley Additive exPlanations, based on game-theory **Shapley values**: each feature's fair contribution to the output.

> **BANGLA INTUITION**
> ফুটবল দল জিতলে কোন খেলোয়াড়ের অবদান কতটা? Shapley value প্রত্যেকের "ন্যায্য অবদান" হিসাব করে। SHAP একইভাবে প্রতিটি feature এর অবদান বের করে।

- **Method:** SHAP `GradientExplainer` (= Integrated Gradients + SmoothGrad), v0.42.1, modified for multi-omic input.
- **Global importance (latent space):** array shape (N_{latent}, $N_{samples}$, $N_{features}$) → take **absolute value** → **sum over latent dims** → **average over samples** → one importance per feature.
- **Reconstruction importance (e.g. drug):** shape (N_{drugs}, $N_{samples}$, $N_{features}$) → average over samples only → importance of each feature *per drug*.
- Subsampling for compute: 20% of samples (drug, copy number), only 20 samples for the huge omics.

Findings (আবিষ্কার)

- Highest average importance by omic: **metabolomics, drug response, copy number**.
- Conditionals: average modest, but **TP53 mutation, growth rate, haematopoietic/lymphoid tissue** rank very high.
- Recovers known biology: **CDKN2A** copy number loss, sensitivity to SRC inhibitor **dasatinib**.
- Top metabolite: **1-methylnicotinamide** (nicotinate/nicotinamide metabolism).

7. Key Discovery: 1-methylnicotinamide, EMT, Drug Resistance

EMT = Epithelial-to-Mesenchymal Transition: stationary, tightly attached epithelial cells (স্থির কোষ) become mobile mesenchymal cells (চলমান কোষ). Linked to **drug resistance** (ওষুধ প্রতিরোধ) and **metastasis** (ক্যান্সার ছড়ানো).

$$\text{NNMT enzyme} \uparrow \rightarrow \text{1-methylnicotinamide} \uparrow \rightarrow \text{EMT state (VIM, CDH1 markers)} \rightarrow \text{drug resistance}$$

- Drugs with highest SHAP importance from this metabolite: Daraprim (pyrimethamine), UNC0638, Entinostat, BIX02189, Methotrexate. Four of the top five are known EMT-related (UNC0638, Entinostat, BIX02189 suppress EMT; methotrexate induces it).
- **Daraprim** was not in the dataset as an anticancer drug, so it is a **drug repurposing** (পুরনো ওষুধের নতুন ব্যবহার) candidate. Other top features for it: GPX1 protein, KRAS, KMT2D (both EMT-linked).
- Consistent with PC-9 lung cancer line (EGFR mutant) and A549 EMT studies.

> **CAUTION**
> Computational evidence only. The authors state that **experimental validation** (পরীক্ষামূলক যাচাই) is still needed.

8. Limitations and Future Work (সীমাবদ্ধতা)

- **Small sample size** (~1,500 lines): more complex VAE designs gave *worse* reconstruction.
- **Proteomics** reconstruction weaker due to intrinsic sparsity.
- **MNAR** (Missing Not At Random): values are missing for a reason (e.g. low-abundance proteins), not randomly; not yet handled.
- SHAP has theoretical limits; biological meaning of some top features is hard to verify.
- Only **cell lines**, not patient data. Future: organoids, PDXs, **transfer learning**, imaging and text modalities (multi-modal augmentation).

9. Training Setup (Table 1)

Hyperparameter	Value	Hyperparameter	Value
Epochs	500	Joint latent dims	200
Batch size	256	Dropout	0.4
Learning rate	3e-4 (Adam)	View dropout	0.3 (text: 0.5)
KL weight λ	1e-4	Contrastive weight α	0.005
Activation	PReLU	Weight decay	5e-4
Scheduler	ReduceLROnPlateau (factor 0.6, patience 7)	Tuning	Optuna, 600 trials, 150 epochs each

Preprocessing: z-score all datasets except copy number; missing values set to 0 and **masked from loss**; features with >85% missing dropped; CRISPR scaled so median essential = −1 and non-essential = 0. Split stratified by haematopoietic/lymphoid lines (suspension vs adherent).

10. Quick Q&A for Exam (দ্রুত প্রশ্নোত্তর)

Question	Answer
What type of model is MOSA?	Unsupervised conditional multi-view VAE with late integration, asymmetrical design, view dropout, contrastive loss.
Why is view dropout needed?	Prevents large omics from dominating and forces learning cross-omic relationships, improving generalisation and explainability.
Why asymmetrical?	Low-variance features add no information; dropping them from the encoder cuts parameters by 39.2% with no loss in reconstruction.
Why use conditionals?	Genetic background and tissue influence every omic, so they contextualise inputs and let the decoder generate condition-specific data.
What are the 3 loss terms?	MSE reconstruction + weighted KL divergence + contrastive (tissue) loss.
How was generalisation proven?	Unseen GDSC drug data (r=0.87), CCLE proteomics, CTD2 drug data, 272 RNA-less lines (r=0.90).
What does SHAP add?	Feature-level explanation of the latent space and reconstructions; led to the 1-methylnicotinamide to EMT hypothesis.
MSE vs Pearson r?	MSE = absolute size of error (lower better, scale-dependent). r = linear agreement/trend (higher better, scale-free).
Why MOSA over MOFA?	Slightly better numbers (0.87 vs 0.86), but main gain is non-linearity, conditionals, view dropout and SHAP explainability.

Glossary (শব্দভান্ডার)

Term	বাংলা	Term	বাংলা
Imputation	হারানো মান অনুমান করে বসানো	Latent space	লুকানো সংক্ষিপ্ত স্থান
Augmentation	ডেটা বৃদ্ধি / সম্পূরণ	Reconstruction	পুনর্গঠন
Biomarker	জৈব সূচক	Gene essentiality	জিনের অপরিহার্যতা
Cross-validation	ডেটা ভাগ করে বারবার পরীক্ষা	Interpretability	ব্যাখ্যাযোগ্যতা
Sparsity	বিরলতা (অনেক ফাঁকা মান)	Heterogeneity	বৈচিত্র্য / অসমজাতীয়তা